Bashar Kabbarah

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Education

University of California, Berkeley — Expected May 2029
B.S. Electrical Engineering and Computer Sciences; GPA: 3.9/4.0 — *Berkeley, CA*

- Regents' and Chancellor's Scholar | SEED Honors Scholar | National Merit Scholar
- **Publication:** co-author, *Journal of Emerging Investigators* (Jan. 2026), deep learning for cervical spine X-ray classification at **95.73%** accuracy; presented at SCCUR.

Experience

Software Engineer — Aug. 2026 – Present
Recursive Self Improvement Labs | literati.ai — *Remote*

- Built and benchmarked the proprietary **real-time compilation engine** behind the platform's live preview, adding **streaming page delivery** to render pages **∼20× faster** and cut preview updates from 3–15 s to **under 2 s**.
- Designed the correctness benchmark gating every engine release: **11,000+** edits modeled on real revision histories of **300 academic papers**, verified byte-for-byte against **∼40,000** reference compiles across a reproducible **6-machine** fleet.
- Surfaced and drove to fix a **critical cross-tenant data-isolation flaw** pre-launch, plus rendering defects invisible to pixel-difference testing; certified every release **byte-identical** to reference under gated rollouts.

Undergraduate Researcher — June 2026 – Aug. 2026
MIT Media Lab | Multisensory Intelligence Group — *Cambridge, MA*

- Extended a **trimodal contrastive** pipeline (**InfoNCE**, **DINOv3 ViT-B/16**, **biGRU**) over tactile, video, and **MANO** pose data, showing touch alone predicts finger-motion direction **533 ms ahead** at **AUC 0.60–0.69** (0.50 chance).
- Proved the signal **not recoverable from hand kinematics** (**+0.039 AUC** over raw pose baseline, 95% CI excludes zero) with a causal linear probe, participant-disjoint splits, and **36 pre-registered tests**.
- Raised cross-modal retrieval **mAP 2.7**× (**4.4**× on unseen participants) by swapping pooled for sequence encoders; traced a standing null result to the target, not the model: whole-hand rotation accounts for **∼95%** of it, masking finger motion.
- First-author poster, MIT Research Symposium; contributing author on the group's **ICLR** submission.

Machine Learning Engineering Intern — Jan. 2026 – June 2026
BFAI Semiconductor Solutions — *Nagoya, Japan*

- Led production **AI defect inspection systems** for Tier-1 Japanese manufacturers (Ibiden, Canon, Kioxia), building **object detection**, **classification**, and **segmentation** models in **PyTorch**.
- Designed **real-time inference pipelines** across **∼20 inspection units** with coordinate transformation for robotic integration; ran **data augmentation** R&D (**SMOTE**, generative) for limited defect data.

Software Engineering Intern — Aug. 2025 – Dec. 2025
UC Berkeley DNA Sequencing Facility — *Berkeley, CA*

- Automated an Oxford Nanopore amplicon pipeline with a **Bash/Python** scheduler, eliminating manual per-barcode execution and cutting operator time **95%** across **96-barcode plates** for **20+ customers daily**.
- Architected **SLURM** job orchestration on Savio **HPC** with parallel submission and dynamic resource allocation.
- Integrated a **Nextflow** workflow for consolidated HTML reporting and refactored the **pandas** pipeline for grouped outputs.

Projects

Machine Olfaction Rig, MIT Media Lab | *KiCad, ESP32, Embedded C++, I2C/UART, Python* — Summer 2026

- Architected and built a **25-sensor gas-sensor array** (electronic nose) on **ESP32** identifying substances from their vapor signatures, streaming **20+ analog channels** over dual **I2C** buses into a Python acquisition pipeline.
- Trained and benchmarked **7 classifier families** (**ExtraTrees**, **SVM**, LDA, logistic regression) on physics-informed transient features, hitting **92% accuracy** on 5-way substance ID under leave-one-day-out CV over **205 runs**; consensus abstention lifts this to **97%** at 86% coverage.
- Extended the rig's electrical design into a fabricated two-board **PCB** in **KiCad**, adding dedicated power regulation and reverse-polarity protection for a more robust sensor stack under single-sided CNC-milling **DFM** constraints.

Technical Skills

Languages: Python, C/C++, Java, SQL, Bash, Scheme, HTML/CSS, x86 Assembly, RISC-V, LaTeX
Frameworks & Tools: PyTorch, scikit-learn, NumPy, pandas, FastAPI, React, Git, Unix, SLURM, Docker, KiCad
ML/AI: Deep Learning, Contrastive & Multimodal Learning, Object Detection, Ensemble Methods, Time-Series Classification
Systems & Hardware: Embedded (ESP32), PCB Design & DFM, I2C/UART, Signal Processing, Benchmark Design, HPC
Courses: Data Structures, Discrete Math & Probability, Machine Structures, Linear Algebra, Data Science; CS189 (Machine Learning) and CS170 (Algorithms) in progress